Demo_7_Transformations_2025

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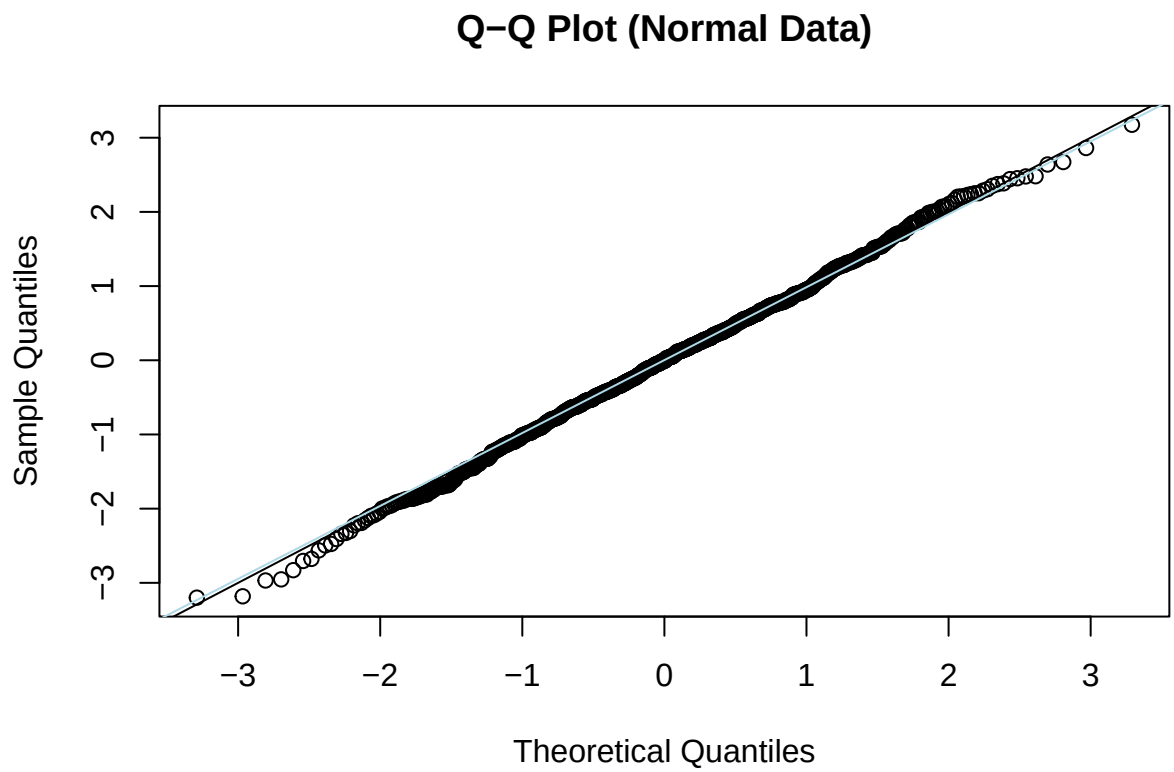
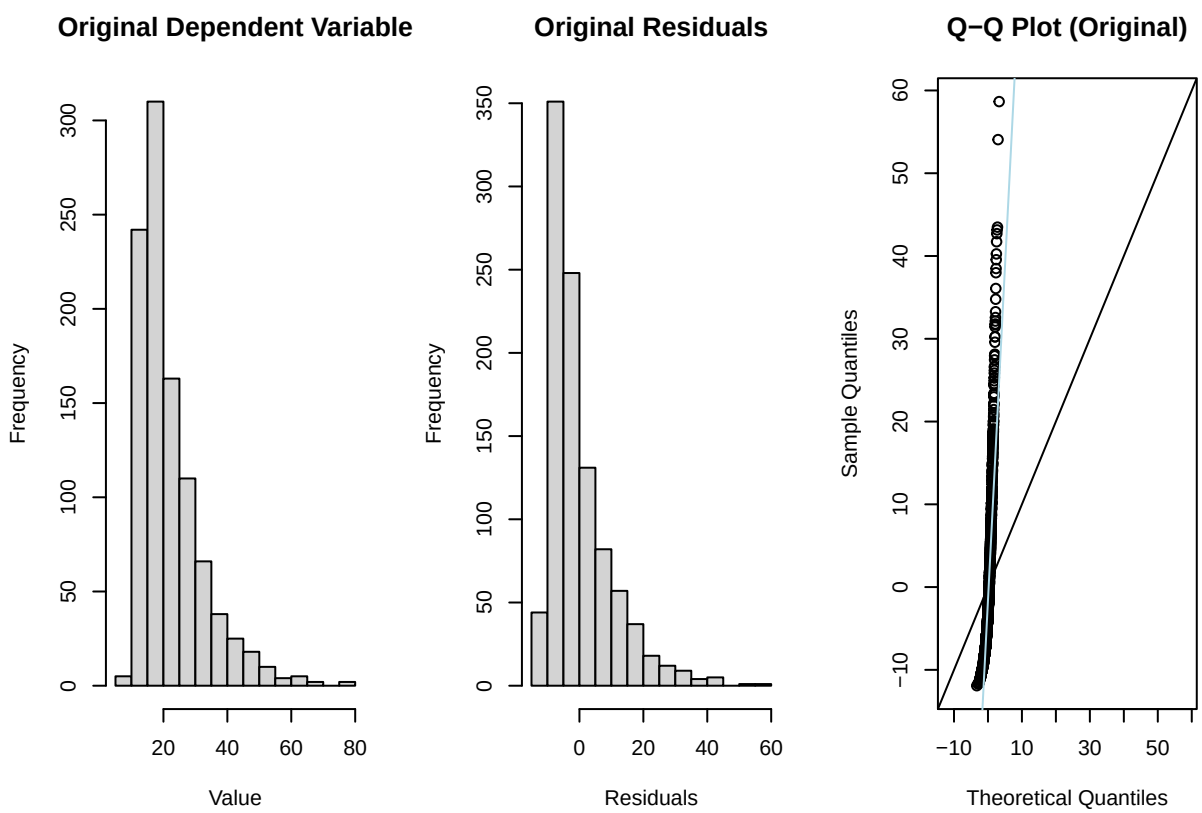
```r
knitr::opts_chunk$set(echo = FALSE)
knitr::opts_chunk$set(message = FALSE)
knitr::opts_chunk$set(warning = FALSE)
library(tinytex)

# The following two functions are copies form GenABEL package by Y. Aulchenko,
# which are very useful for transformations, but not available from CRAN or
# Bioconductor for modern R. We include them as part of the script.
# The ztransform function standardizes the residuals of a linear model,
# while the rntransform function performs a rank-based inverse normal transformation
# on the dependent variable.
# Load the R script from GitHub
source("https://raw.githubusercontent.com/CrisVanHout-LIIGH/Class/main/Demo_7_Transformations/RINT.r")

# Simulate example data
set.seed(42)
n <- 1000
genotype <- sample(c("0", "1", "2"), size = n, replace = TRUE, prob = c(0.25, 0.5, 0.25))
dependent <- ifelse(genotype == "0", 1, ifelse(genotype == "1", 2, 3)) + 100*rexp(n, rate=10) + rnorm(n

# Create a data frame
data <- data.frame(genotype = as.numeric(genotype), dependent = dependent)

# Inspect residuals for normality
model_original <- lm(dependent ~ genotype, data = data)
residuals_original <- residuals(model_original)


# Apply a transformation (natural log transformation)
data$dependent_transformed <- log(data$dependent)
model_transformed <- lm(dependent_transformed ~ genotype, data = data)
residuals_transformed <- residuals(model_transformed)

# Set common axis limits based on the range of all residuals
common_limits <- range(c(residuals_original, residuals_transformed))

# Reminder of interpretation of qq plot: If the data is normally distributed,
# the points in the Q-Q plot should fall approximately along a straight line.
# If the data is not normally distributed, the points will deviate from the straight line.

#example
par(mfrow = c(1, 1))
```

```r
normal_data <- rnorm(1000)
qqnorm(normal_data,main="Q-Q Plot (Normal Data)")
abline(0,1)
qqline(normal_data, col = "lightblue")
```



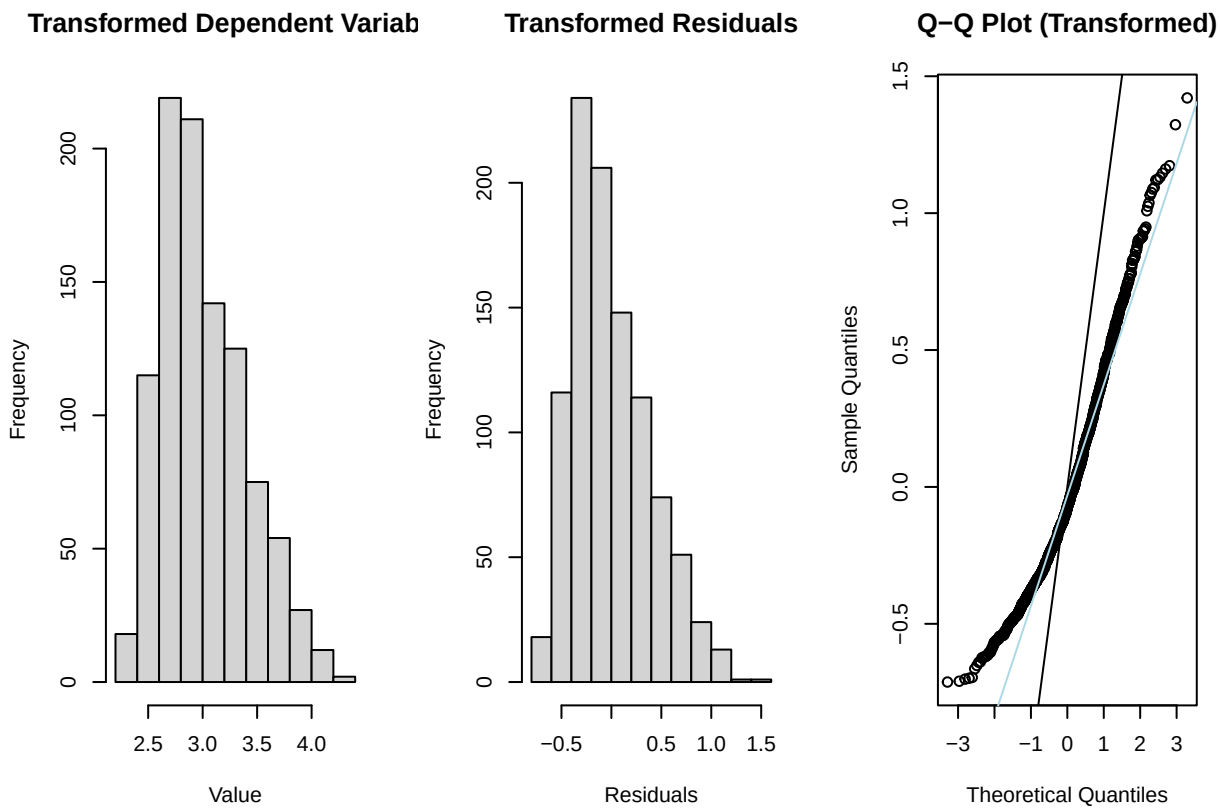

```r
# Purpose of the Fitted vs. Residuals Plot:
# Assess Linearity:
# This plot helps determine if the relationship between predictors and the dependent
# variable is linear. Ideally, the residuals should scatter randomly around zero across
# the range of fitted values. If you see patterns (e.g., curves), it may indicate
# non-linearity, suggesting the need for a transformation of the dependent variable
# or predictors.

# Detect Heteroscedasticity:
# Unequal variance (heteroscedasticity) appears as a funnel shape or clustering in
# the plot. Such patterns suggest that the variance of the errors changes with the
# fitted values, which violates a key assumption of linear regression.
#Transformations can address this issue.

# Identify Outliers or Influential Points:
# Residuals that deviate substantially from the zero line or clusters of outliers
# in specific regions of the plot can indicate problems with certain data points.
# RINT or similar transformations can help mitigate their impact by reducing the
# influence of extreme values.
```

```r
# Diagnose Model Fit:
# If the residuals show clear systematic trends, the model is likely mis-specified
# (e.g., missing interactions or non-linear relationships).

# Connection to Transformations:
# If the plot exhibits patterns or non-random residuals, this signals a need for
#corrective measures such as transformations. For example:

# A log transformation can stabilize variance and linearize relationships.
# A RINT transformation focuses on normalizing residuals and is especially
# helpful when extreme values are a concern.

# Let's look at the residuals of the original and transformed models:

par(mfrow = c(1, 3))
hist(dependent, main = "Original Dependent Variable", xlab = "Value")
hist(residuals_original, main = "Original Residuals", xlab = "Residuals")
qqnorm(residuals_original, main = "Q-Q Plot (Original)", xlim = common_limits, ylim = common_limits)
abline(0,1)
qqline(residuals_original,col="lightblue")
```

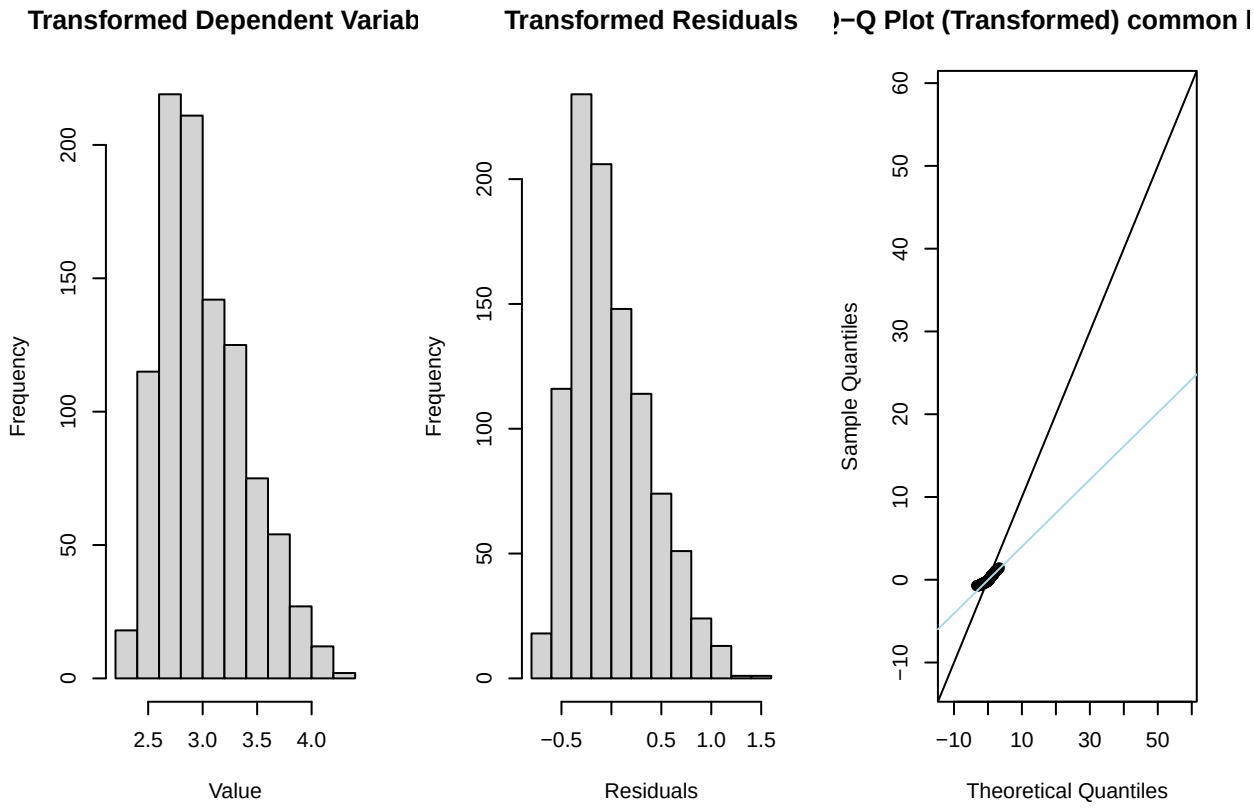



```r
par(mfrow = c(1,3))
hist(data$dependent_transformed, main = "Transformed Dependent Variable", xlab = "Value")
hist(residuals_transformed, main = "Transformed Residuals", xlab = "Residuals")
qqnorm(residuals_transformed, main = "Q-Q Plot (Transformed)")
```

```r
abline(0,1)
qqline(residuals_transformed,col="lightblue")
```




```r
par(mfrow = c(1,3))
hist(data$dependent_transformed, main = "Transformed Dependent Variable", xlab = "Value")
hist(residuals_transformed, main = "Transformed Residuals", xlab = "Residuals")
qqnorm(residuals_transformed, main = "Q-Q Plot (Transformed) common limits", xlim = common_limits, ylim
abline(0,1)
qqline(residuals_transformed,col="lightblue")
```




```r
# Fitted vs. Residuals: Original Model with Jittered Fitted Values
plot(jitter(fitted(model_original)), residuals(model_original),
     main = "Untransformed Model (Jittered Fitted)",
     xlab = "Fitted Values (Jittered)",
     ylab = "Residuals",
     pch = 20, col = "blue")
abline(h = 0, col = "red", lwd = 2)

# Fitted vs. Residuals: Log-Transformed Model with Jittered Fitted Values
plot(jitter(fitted(model_transformed)), residuals(model_transformed),
     main = "Log-Transformed Model (Jittered Fitted)",
     xlab = "Fitted Values (Jittered)",
     ylab = "Residuals",
     pch = 20, col = "green")
abline(h = 0, col = "red", lwd = 2)

# Compare the summaries
summary(model_original)
```

```
##
## Call:
## lm(formula = dependent ~ genotype, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.915  -7.046  -3.202   3.819  58.655
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.1473     0.5543  38.152   <2e-16 ***
## genotype      0.8756     0.4627   1.892   0.0587 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.06 on 998 degrees of freedom
## Multiple R-squared:  0.003575,   Adjusted R-squared:  0.002577
## F-statistic: 3.581 on 1 and 998 DF,  p-value: 0.05874
```

```r
summary(model_transformed)
```

```
##
## Call:
## lm(formula = dependent_transformed ~ genotype, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.71257 -0.30333 -0.07418  0.24186  1.42088
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.95867    0.02150 137.584  < 2e-16 ***
## genotype     0.05023    0.01795   2.798  0.00524 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3903 on 998 degrees of freedom
## Multiple R-squared:  0.007783,   Adjusted R-squared:  0.006789
## F-statistic: 7.828 on 1 and 998 DF,  p-value: 0.005242
```

```r
# Shapiro-Wilk Test The W statistic ranges between 0 and 1.
# A value close to 1 indicates that the data is nearly normal,
# while smaller values suggest deviations from normality

shapiro.test(residuals_original)   # Before transformation
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals_original
## W = 0.83201, p-value < 2.2e-16
```

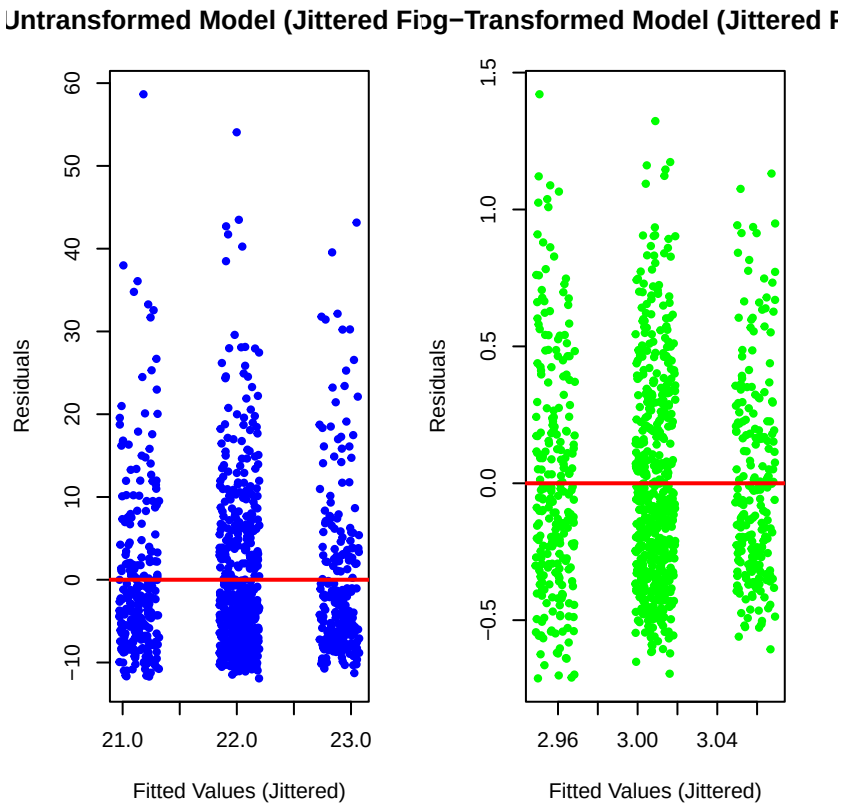
```r
shapiro.test(residuals_transformed)  # After transformation
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals_transformed
## W = 0.95718, p-value < 2.2e-16
```

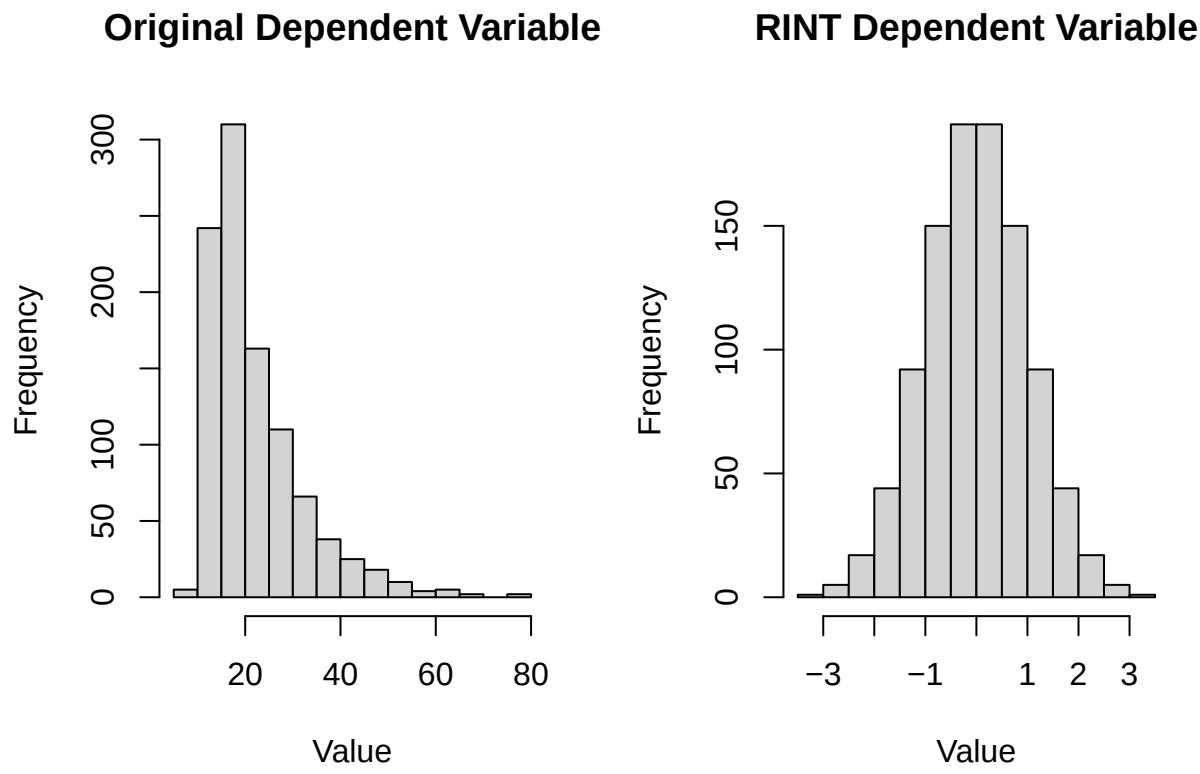
```r
# Interpret the beta coefficients
# The beta coefficient for genotype=1 is the expected difference in
# the dependent variable between genotype=1 and genotype=0, holding
# all other variables constant.
# When the dependent variable is log-transformed (log(y)), the beta
# coefficient represents the expected change in the logarithm of the
# dependent variable (log(y)) per unit change in the predictor variable
# (e.g., genotype levels 0,1,2).
# To interpret the beta coefficient in terms of the original (unlogged)
# dependent variable, you can exponentiate it using exp(beta).

# The resulting value gives the multiplicative effect of the predictor
# on the dependent variable. For example:
# If exp(beta) > 1, the predictor increases the dependent variable
# by a factor of exp(beta).
# If exp(beta) < 1, the predictor decreases the dependent variable
# by a factor of exp(beta).
# If exp(beta) = 1, the predictor has no effect on the dependent variable.
# After transformation, the beta coefficient reflects relative changes
# instead of absolute changes in the dependent variable.

# Suppose the beta coefficient for a genotype in the transformed model is
# 0.05023:

# Log Scale: For each unit change in the predictor (genotype), the log of
# the dependent variable increases by 0.05023.
# Exponentiated Scale:

exp(model_transformed$coefficients[[2]])
```

```
## [1] 1.05151
```

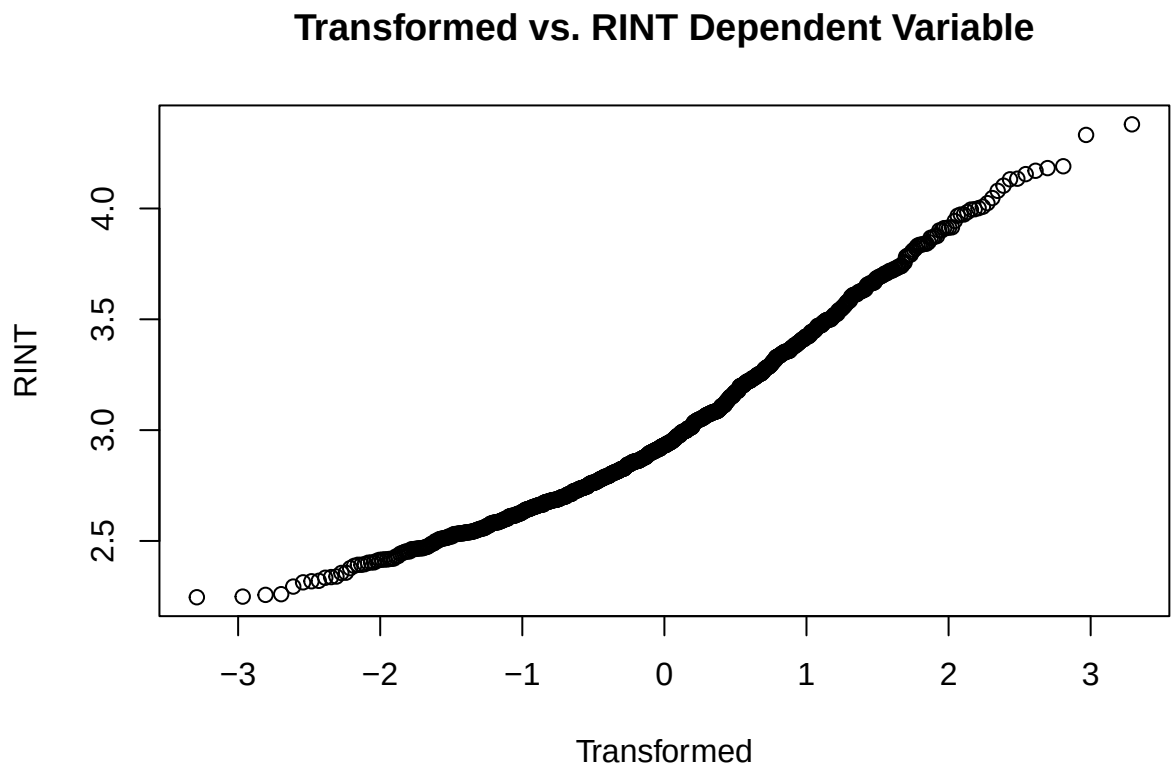
```r
# Indicating that the dependent variable increases by approximately 5.1%
# per unit change in the predictor.
# By transforming back to the original scale, you make the interpretation
# more intuitive and relevant to the practical context.

# Evaluate another type of transformation, the rank-based inverse normal
# transformation (RINT) The RINT is a non-parametric transformation that
# converts the dependent variable into a standard normal distribution.
# The RINT maps the ranked values of the dependent variable onto a standard
# normal distribution, effectively normalizing the variable while preserving
# its rank order. This transformation is useful when the residuals of the
# model are not normally distributed, even after applying other transformations.

data$dependent_rint <- rntransform(data$dependent)

par(mfrow = c(1,2))
```



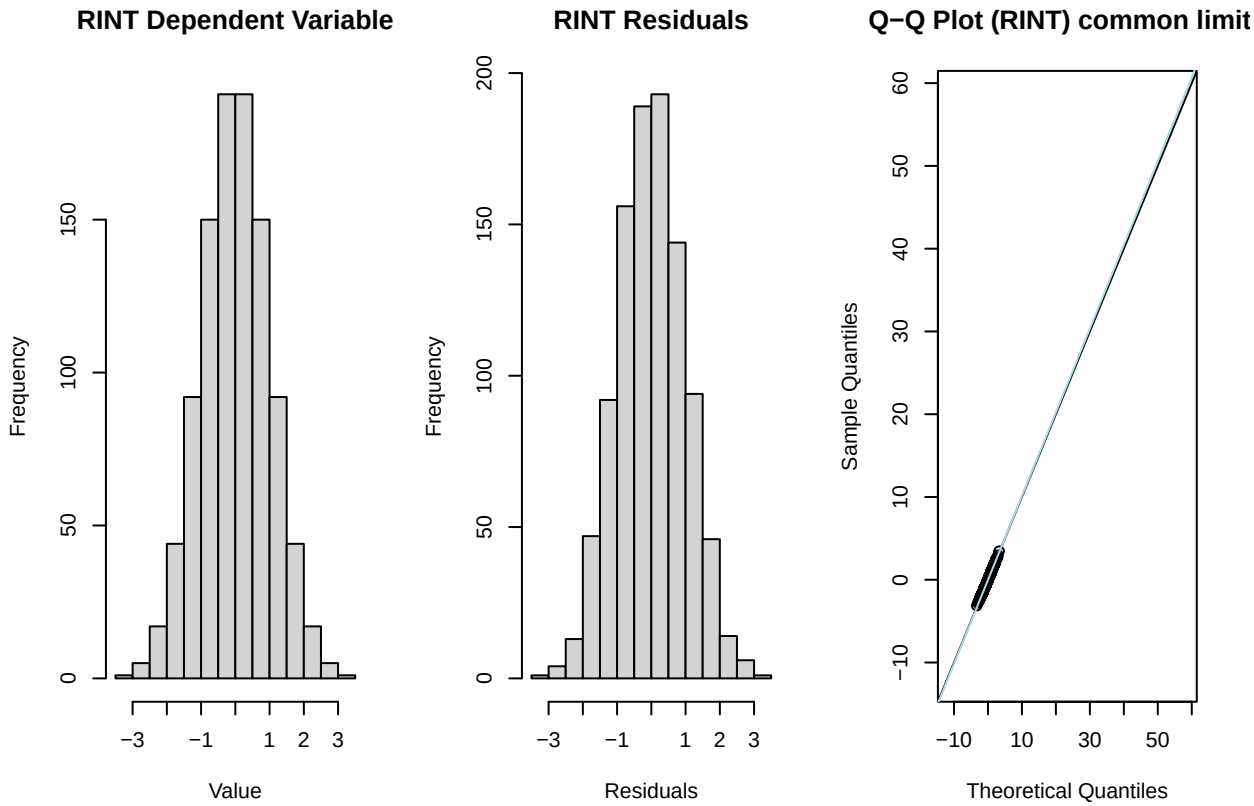

```r
hist(data$dependent, main = "Original Dependent Variable", xlab = "Value")
hist(data$dependent_rint, main = "RINT Dependent Variable", xlab = "Value")
```




```r
par(mfrow = c(1,1))
plot(data$dependent_rint, data$dependent_transformed, main = "Transformed vs. RINT Dependent Variable",
```




```
# Fit the model with RINT-transformed dependent variable
model_rint <- lm(dependent_rint ~ genotype, data = data)
residuals_rint <- residuals(model_rint)

par(mfrow = c(1,3))
hist(data$dependent_rint, main = "RINT Dependent Variable", xlab = "Value")
hist(residuals_rint, main = "RINT Residuals", xlab = "Residuals")
qqnorm(residuals_rint, main = "Q-Q Plot (RINT) common limits", xlim = common_limits, ylim = common_limi
abline(0,1)
qqline(residuals_rint,col="lightblue")
```




```r
shapiro.test(residuals_transformed)  # After RINT
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals_transformed
## W = 0.95718, p-value < 2.2e-16
```

```r
#What is the p-value of the Shapiro-Wilk test for the RINTed dependent variable?
#shapiro.test(data$dependent_rint)

# Compare the summaries
summary(model_rint)
```

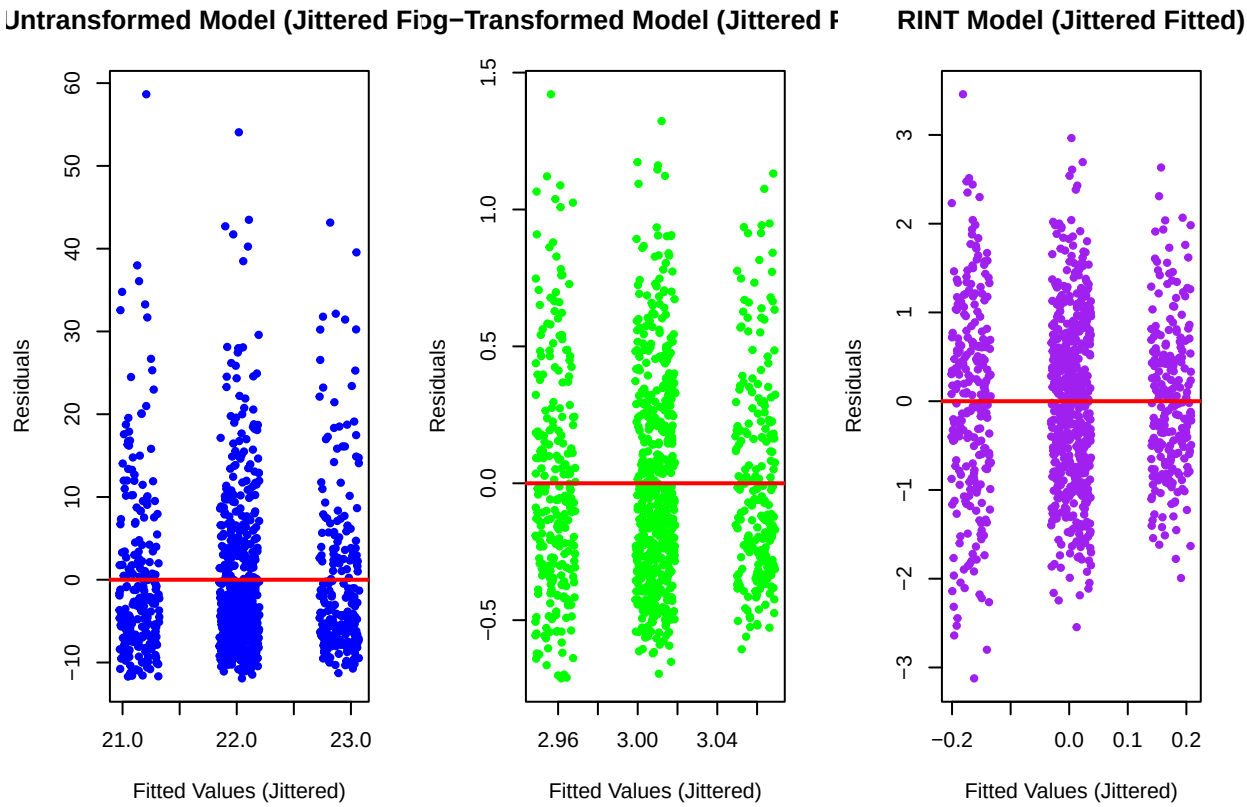
```
##
## Call:
## lm(formula = dependent_rint ~ genotype, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1229 -0.7057 -0.0095  0.6595  3.4581
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.16762    0.05474  -3.062 0.002256 **
```

```
## genotype     0.17086    0.04569   3.739 0.000195 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9934 on 998 degrees of freedom
## Multiple R-squared:  0.01382,    Adjusted R-squared:  0.01283
## F-statistic: 13.98 on 1 and 998 DF,  p-value: 0.0001951
```

```r
# Since RINT normalizes the dependent variable, the beta coefficient reflects
# the standardized effect size. It tells you how much the dependent variable
# (on the normalized scale) changes per unit change in the predictor.
# The values of the beta coefficients cannot be directly translated back to
# the original scale of the dependent variable. They are relative measures
# indicating the direction and magnitude of association within the transformed
# (ranked) scale. RINT reduces the influence of outliers and emphasizes rank
# order rather than raw values. Therefore, the beta coefficient is less sensitive
# to extreme values and primarily describes the overall trend or relationship.

model_rint$coefficients[[2]]
```

```
## [1] 0.1708648
```

```r
#This means that for this genotype, the dependent variable's rank-transformed
# value is expected to increase by 0.17 standard deviations for each unit
# of the predictor variable.

# Compare the residuals of the original, log-transformed, and RINT models
par(mfrow = c(1, 3))

# 1. Fitted vs. Residuals: Original Model with Jittered Fitted Values
plot(jitter(fitted(model_original)), residuals(model_original),
     main = "Untransformed Model (Jittered Fitted)",
     xlab = "Fitted Values (Jittered)",
     ylab = "Residuals",
     pch = 20, col = "blue")
abline(h = 0, col = "red", lwd = 2)

# 2. Fitted vs. Residuals: Log-Transformed Model with Jittered Fitted Values
plot(jitter(fitted(model_transformed)), residuals(model_transformed),
     main = "Log-Transformed Model (Jittered Fitted)",
     xlab = "Fitted Values (Jittered)",
     ylab = "Residuals",
     pch = 20, col = "green")
abline(h = 0, col = "red", lwd = 2)

# 3. Fitted vs. Residuals: RINT Model with Jittered Fitted Values
plot(jitter(fitted(model_rint)), residuals(model_rint),
     main = "RINT Model (Jittered Fitted)",
     xlab = "Fitted Values (Jittered)",
     ylab = "Residuals",
     pch = 20, col = "purple")
abline(h = 0, col = "red", lwd = 2)
```




```
# In Summary
#If the residuals in a linear model do not meet the assumption of normality,
# it can lead to various consequences.

# Inference Issues:
# P-values and Confidence Intervals: Non-normal residuals can make the
# standard errors unreliable, leading to inaccurate p-values and confidence
# intervals for the regression coefficients.

# Type I/II Errors: Increased risk of falsely rejecting the null hypothesis
# (Type I error) or failing to detect a true effect (Type II error).

# Violation of Assumptions:
# While the normality of residuals is not required for unbiased coefficient
# estimation, it is critical for valid hypothesis testing and for relying on
# outputs such as t-tests and F-tests.

# Loss of Predictive Power:
# Non-normal residuals may indicate issues like heteroscedasticity or
# non-linearity, which can reduce the model's predictive accuracy.

# Underestimation of Extreme Values:
# Non-normal residuals, especially when skewed, can result in poor
# predictions for extreme observations (outliers) in the dependent variable.

# Inefficient Parameter Estimation:
```

```r
# Although linear regression is robust to slight deviations from normality,
# severe non-normality can lead to inefficient parameter estimates,
# particularly if the sample size is small.

# To address these issues, you can apply transformations to the dependent
# variable to improve the normality of residuals! The choice of transformation
# depends on the distribution of the dependent variable and the nature of the
# relationship with the predictor variables.

# Using the fitted vs. residuals plot, Q-Q plot, and Shapiro-Wilk test, you
# can assess the normality of residuals and determine whether a transformation
# is necessary.
```